Eightfold AI – Multi-Source Candidate Data Transformer

Technical Design

Pipeline Overview

The solution follows a modular data pipeline:

Recruiter CSV │ ▼ CSV Parser

Resume TXT │ ▼ Resume Parser │ ▼ Data Normalizer │ ▼ Data Merger │ ▼ Data Validator │ ▼ Data Projector │ ▼ Final JSON Output

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Canonical Candidate Model

The pipeline transforms all source data into a single canonical Candidate model.

Each field stores:

- Value
- Confidence Score
- Source Provenance

This enables traceability and conflict resolution across multiple data sources.

Normalization Strategy

The normalization stage standardizes candidate information before merging.

Examples:

- Emails → lowercase
- Phone Numbers → E.164 format
- Names → Title Case
- Skills → duplicate removal and canonical formatting

Merge Strategy

The merger combines structured and unstructured data into one candidate profile.

Rules:

- If both sources contain the same value:
    - Confidence = 1.0
    - Sources = CSV + Resume

- If values differ:
  - Resume value is preferred.
  - Confidence = 0.8
  - Provenance records both sources.
- If only one source contains a value:
  - Confidence = 0.7

Validation

Before exporting:

- Required fields must exist.
- Email format is validated.
- Phone number format is validated.
- Empty skills are rejected.

Configurable Output

The projector reads a runtime JSON configuration.

The configuration controls:

- Included fields
- Field names
- Confidence output
- Provenance output

No Python code changes are required.

Edge Cases

The pipeline handles:

- Missing CSV
- Missing Resume
- Missing fields
- Duplicate skills
- Invalid phone numbers
- Empty values

Assumptions

- Recruiter CSV follows expected column names.
- Resume is available as text.
- Phone numbers are normalized using the Indian region.